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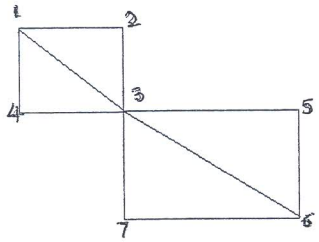
SCHOOL OF TECHNOLOGY MANAGEMENT

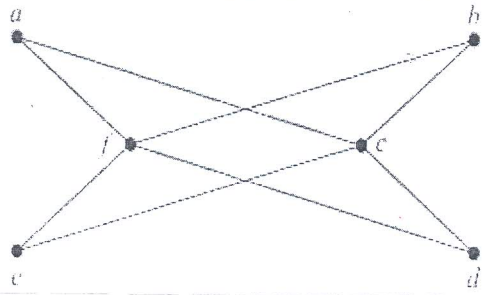
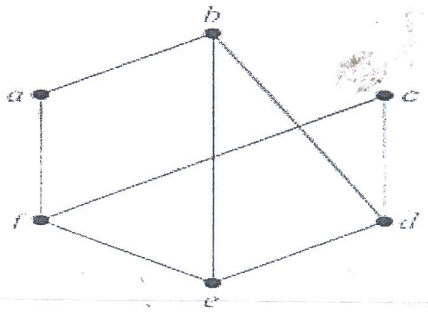
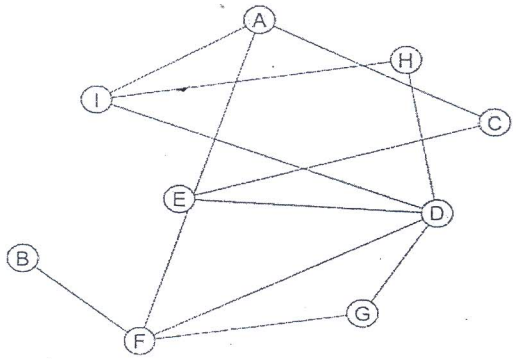
Academic Year: 2023-2024

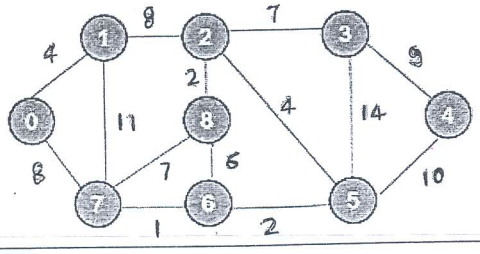
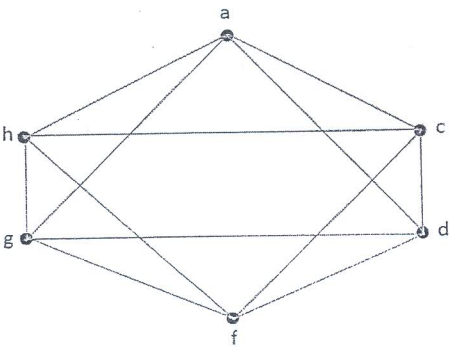
Programs: B. Tech / MBA Tech Streams: DS/AIDS/AIML/AI/CSE(CS)/CS/CE/ETE/CSEDS/CSBS	Year: 2nd /3rd Semester: 3rd /5th
Subject: Discrete Mathematics	Time: 3 hrs. (10 am to 1 pm)
Date: 07/02/2024	No. of Pages: 3
Marks: 100	
~~Final Examination~~/Re-Examination (2021-22, 2022-23)	

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

CO-1 ; SO- 2; BL-3	Q.1 a)	Out of 350 candidates who appear in an examination, it was revealed that 150 failed in calculus and 105 in physics. 35 failed in both (calculus and in Physics). Find how many candidates (i) Pass in both (ii) Pass only in calculus.	4
CO-2 ; SO- 1; BL-2	b)	Show that the statement $(p \vee q) \rightarrow (p \wedge q)$ is a contingency.	4
CO-1 ; SO- 1; BL-1	c)	Given 12 different 2-digit numbers, show that one can choose two of them so that their difference is a two-digit number with identical first and second digit.	4
CO-4 ; SO- 1; BL-3	d)	Does the following graph have an (i) Euler's path (ii) Euler's Circuit? Justify your answer. 	4
CO-4 ; SO-2 ; BL-5	e)	If $\mathbb{Z}_4 = \{0,1,2,3\}$ and $a +_4 b = r$, where r is reminder when we divide $a + b$ by 4. Prove that $(\mathbb{Z}_4 , +_4)$ is a group.	4
CO- 3; SO- 2;	Q.2 a)	Determine the solution for recurrence relation $2a_{n+2} - 11a_{n+1} + 5a_n = 0$ for $n \geq 0$ where $a_o = 2$ and $a_1 = -8$.	6

BL-5			
CO- 1; SO- 2; BL-3	b)	Are the following graphs displayed below bipartite? Justify your answer.   	6
CO-3 ; SO-2 ; BL-4	c)	Let $\mathbb{Z}$ denotes the set of integers. Define a relation R on $\mathbb{Z}$ by setting xRy if and only if $x \equiv y \pmod 6$. Examine whether R is equivalence relation or not. Also find the equivalence classes.	8
CO-2 ; SO- 1; BL-2	Q.3 a)	Show that $2^{2n} - 1$ is divisible by 3 using the principles of mathematical induction.	6
CO- 2; SO- 2; BL-6	b)	By using Welsh Powell graph colouring algorithm, find the chromatic number of the following graph.  	6
CO- 1; SO- 1; BL-4	c)	Let $A = \{1,2,3,4\}$. Examine whether the relation $R = \{(1,2), (1,3), (1,4), (2,3), (2,4), (3,4)\}$ is symmetric, asymmetric, antisymmetric and transitive. Give argument in support of your answer.	8
CO- 3; SO- 2; BL-4	Q.4 a)	Show that the set of non-zero real numbers $(\mathbb{R}\backslash\{0\})$ form an abelian group with respect to binary operation '$*$' defined as $a * b = (a \times b)/2,\ \forall a, b \in \mathbb{R}\backslash\{0\}$	6
CO-2 ; SO-2 ; BL-2	b)	Discuss the satisfiability and validity of the following argument: *If you study regularly, they you will score distinction. You study regularly. Therefore, you will not score distinction.*	6
CO- 3; SO- 1; BL-5	c)	Determine the solution of the following recurrence relation: $a_n - 4\, a_{n-1} = 2 \cdot 4^n$, where $n \geq 1$; *and* $a_0 = 3$.	8
CO- 2; SO- 1; BL-4	Q.5 a)	Prove that the additive groups $G = (\mathbb{Z}, +)$ and $H = (2\mathbb{Z}, +)$ are isomorphic.	6

CO-3 ; SO- 1; BL-4	b)	Name the methods can be used for proving propositional statements? Prove that there is no smallest positive rational number.	6
CO- 3; SO- 1; BL-3	c)	Apply Kruskal's Algorithm to find the minimal spanning tree of the weighted graph: 	8
CO-1 ; SO- 2; BL-4	Q.6 a)	If $f: \mathbb{R} \longrightarrow \mathbb{R}$, $g: \mathbb{R} \longrightarrow \mathbb{R}$ is defined as $f(x) = 7x + x^2$, $g(x) = -7x^3$. Determine the composition functions $g \circ g$, $f \circ g$ and $g \circ f$.	6
CO- 4; SO- 1; BL-3	b)	Define the planar graph and determine whether the following graph is Planar. 	6
CO- 4; SO- 1; BL-6	c)	If the addition and multiplication modulo 7 is defined on the set $R = \{0, 1, 2, 3, 4, 5, 6\}$ then show that algebraic structure $(R, +_7, \times_7)$ is a ring with unity. Is it a field?	8
CO- 2; SO- 2; BL-3	Q.7 a)	For the given prepositions, identify which are equivalencies or tautologies. i) $p \wedge \sim q$ ii) $\sim p \vee q$ iii) $\sim (\sim p \vee q)$ iv) $\sim p \vee (p \wedge \sim q)$ v) $q \wedge \sim (\sim p \vee q)$	6
CO-3 ; SO- 7; BL-2	b)	Let $G = \{5, 15, 25, 35\}$ set of congruence classes under multiplication modulo 40 $(\times_{40})$, (i) Construct composition table with respect to '$\times_{40}$'. (ii) Identify Identity element of $(G, \times_{40})$. (iii) Identify the inverse of all the elements under $\times_{40}$.	6
CO- 4; SO- 1; BL-3	c)	i) Construct all connected graphs of four vertices which are not isomorphic. ii) Construct a graph which is Euler circuit but not Hamiltonian circuit and explain your answer.	8